

Net Improvements for the Core Engine

Adoptable elements from two reviewed papers for the ML cross-sectional S&P 500 long-short strategy

Prepared for the ML_Project core engine • May 2026

This note lists **only the elements that would be a net improvement to add** to the existing project plan — nothing that merely restates what is already scoped (return prediction, walk-forward backtesting, transaction costs, sector-neutral construction, the GMM overlay). Each item is drawn from one of the two reviewed papers and mapped to where it plugs into the core engine. The two sources are **Glawion (2021)**, "Quantitative Trading with Machine Learning" (Stanford CS229), a methodologically sound return-prediction and walk-forward study, and **Yu et al. (2025)**, "Gradient Boosting Decision Tree with LSTM for Investment Prediction" (IEEE ACCTCS), from which only the ensemble-architecture ideas are worth borrowing.

1. Nested expanding-window split with a dedicated validation fold

Add an explicit **three-way sequential split** on top of the planned walk-forward: an expanding *training* window, a rolling *validation* window used solely for hyperparameter selection, and a separate *out-of-sample test* step that the model never sees during tuning. After each step the test period is folded back into train/validation and the window rolls forward (Glawion runs 109 such steps: 5-year train, 1-year validation, 1-month test). Splits are sequential, never random. The net gain over a plain walk-forward is that hyperparameters are chosen on data fully disjoint from the evaluation set, which closes a common and easily-overlooked tuning-leakage path and makes the reported Sharpe more defensible.

Source: Glawion (2021), section 4.4 and Figure 2.

2. Add a regularized linear model (Elastic Net) as a first-class contender

Include an **Elastic Net** in the model line-up rather than relying on tree ensembles alone. In Glawion's horse race it produced the lowest test RMSE/MAE at every forecast horizon and beat XGBoost and LightGBM, which is consistent with Gu, Kelly & Xiu's finding that regularized linear models are hard to beat in the low signal-to-noise, highly-correlated feature regime typical of equity panels. It is cheap to fit, gives an honest performance floor, and guards against over-engineering the engine. The L1/L2 mix also handles the heavy cross-sectional correlation among characteristics that defeats a plain lasso.

Source: Glawion (2021), section 4.2.1 and Table 1.

3. Signal-strength entry threshold on predicted returns

Trade a name only when the **magnitude of its predicted return clears a threshold**, and raise that threshold to reduce trading. Glawion shows this lever directly controls turnover — his lowest-threshold configurations fired well over a thousand trades, enough that realistic per-trade costs alone would exceed the bankroll, while higher thresholds achieved comparable or better results with a fraction of the trades. For the core engine this is a concrete turnover- and cost-control mechanism on top of decile ranking; it is also the exact lever the GMM regime layer is meant to modulate later (tighten the threshold in turbulent regimes), so building it now makes the overlay a clean drop-in.

Source: Glawion (2021), section 6 and Table 2.

4. Heterogeneous stacking with out-of-fold meta-learning

As an optional layer above the linear/GBDT baseline, **stack a tree-based learner with a sequence learner** and combine them with a lightweight meta-model, so the engine captures both nonlinear cross-sectional interactions and temporal dynamics. The one critical implementation detail to carry over is that base-learner predictions feeding the meta-model must be generated **out-of-fold within each training window** (cross-validated, never in-sample) — this is what prevents the leakage and overfitting that naive stacking introduces. Keep it secondary to the simpler baseline and adopt only if it earns its keep out-of-sample.

Source: Yu et al. (2025), section II and section III.F (stacking / meta-learner), with leakage-safe out-of-fold construction per Wolpert (1992).

5. Native categorical encoding of GICS sector

Feed the **GICS sector as a native categorical feature** using CatBoost or LightGBM, which handle categoricals directly without one-hot expansion. This lets the model learn sector-conditional effects (a characteristic can predict differently within energy vs. tech) and complements the sector-neutral portfolio construction rather than duplicating it. It is a low-cost feature-engineering addition that the chosen GBDT libraries already support.

Source: Yu et al. (2025), section I and sections III.C-D (CatBoost/LightGBM categorical handling).

6. SHAP-based feature attribution for the GBDT predictions

Add **SHAP attribution** on the gradient-boosted predictions, computed per rebalance period. This validates that the signal is driven by economically sensible characteristics (momentum, value, quality, etc.) rather than artifacts, supports the methodology write-up, and acts as an early-warning check for overfitting when a single feature begins to dominate the predictions.

Source: Yu et al. (2025), section I (feature-importance / SHAP interpretability).

7. Compact macro / credit conditioning block

Augment the firm-level characteristics with a **small block of macro and credit-market features** — e.g. a term spread, a credit spread, short-rate changes, and a volatility measure. Glawion adds an analogous macro/yield-curve/credit set (output gap, CPI, sovereign yield, interbank and swap rates, a credit-spread index with short-horizon changes) to capture the state of the economy. This supplies the model with regime-relevant conditioning information and mirrors the characteristics-times-macro interaction extension in Gu, Kelly & Xiu. Keep the block compact to avoid diluting the cross-sectional signal.

Source: Glawion (2021), section 3 (macroeconomic and credit features).

Summary: where each addition plugs in

Net-improvement to add	Pipeline stage	Source
Nested train / validation / OOS-test expanding window	Backtest protocol	Glawion
Elastic Net as first-class model + benchmark	Model line-up	Glawion

Net-improvement to add	Pipeline stage	Source
Signal-strength entry threshold (turnover / regime lever)	Portfolio construction	Glawion
Heterogeneous stacking, out-of-fold meta-learning	Prediction model (optional)	Yu et al.
GICS sector as native categorical feature	Feature engineering	Yu et al.
SHAP feature attribution	Validation / interpretability	Yu et al.
Compact macro / credit conditioning block	Feature engineering	Glawion

Scope note: items already in the project plan (predicting excess returns, the walk-forward backtest itself, transaction-cost modeling, sector neutrality, and the GMM regime overlay) are intentionally excluded here; only refinements and net-new additions are listed.